

Brutal Paper B50 - Nutrition in Animals, Exponents & Powers, Wastewater Story & Lines & Angles

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A grazing cow makes 6 litres of saliva each hour while it grazes for 7 hours, then only 1.5 litres each hour for the rest of a 24-hour day. How much saliva does it make in the whole day? [\[Nutrition in Animals\]](#)
 - (A) 67.5 L
 - (B) 75 L
 - (C) 51 L
 - (D) 42 L
 - (E) 144 L
 - (F) 60 L
 - (G) 25.5 L

2. Light travels at about 3×10^8 metres per second. How far does it travel in 2×10^{-3} seconds? Give the answer in standard form. [\[Exponents & Powers\]](#)
 - (A) 6×10^5 m
 - (B) 5×10^5 m
 - (C) 3×10^5 m
 - (D) 6×10^{11} m
 - (E) 6×10^6 m
 - (F) 6×10^{-5} m
 - (G) 6×10^{24} m

3. A sewage treatment plant cleans 4.5 million litres of wastewater every day. How many litres does it clean in 30 days? [\[Wastewater Story\]](#)
 - (A) 1.35 million L
 - (B) 13.5 million L
 - (C) 150 million L
 - (D) 1350 million L
 - (E) 135000 L
 - (F) 45 million L
 - (G) 135 million L

4. Two angles are complementary, and one of them is 14 degrees larger than the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)
 - (A) 62 degrees
 - (B) 38 degrees
 - (C) 45 degrees
 - (D) 76 degrees
 - (E) 83 degrees
 - (F) 48 degrees
 - (G) 52 degrees

5. The small intestine is about 6 metres long. The tiny villi on its inner wall increase the absorbing surface 30 times. If the plain tube offers 0.25 square metre of surface per metre of length, what is the total absorbing area with villi? [\[Nutrition in Animals\]](#)
- (A) 7.5 m²
 - (B) 36 m²
 - (C) 45 m²
 - (D) 30 m²
 - (E) 50 m²
 - (F) 1.5 m²
6. Simplify $(2^3 \times 2^4) / 2^5$ as a single power of 2, and then give its value. [\[Exponents & Powers\]](#)
- (A) $2^2 = 8$
 - (B) 2^{-2}
 - (C) $2^2 = 2$
 - (D) $4^2 = 16$
 - (E) $2^2 = 4$
 - (F) $2^7 = 128$
 - (G) $2^{12} = 4096$
7. Of 800 kg of solid waste entering a plant each day, bar screens catch 15% and the grit chamber catches a further 25%. How much waste, in kilograms, passes both to the next stage? [\[Wastewater Story\]](#)
- (A) 200 kg
 - (B) 400 kg
 - (C) 120 kg
 - (D) 240 kg
 - (E) 560 kg
 - (F) 480 kg
 - (G) 320 kg
8. Two supplementary angles are in the ratio 4 : 5. What is the measure of the smaller angle? [\[Lines & Angles\]](#)
- (A) 72 degrees
 - (B) 36 degrees
 - (C) 60 degrees
 - (D) 100 degrees
 - (E) 45 degrees
 - (F) 80 degrees
9. An adult human has 32 teeth: 8 incisors, 4 canines, 8 premolars and 12 molars. What fraction of all the teeth are grinding teeth (premolars and molars together)? [\[Nutrition in Animals\]](#)
- (A) $\frac{2}{3}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{5}{16}$
 - (D) $\frac{5}{8}$
 - (E) $\frac{1}{2}$
 - (F) $\frac{7}{8}$

10. Find the value of $(3^0 + 4^0) \times 5^2$. [\[Exponents & Powers\]](#)

- (A) 2
- (B) 50
- (C) 0
- (D) 75
- (E) 25
- (F) 175

11. In an aeration tank, helpful bacteria reduce 600 kg of organic sludge by three-fifths. How much sludge is left after this? [\[Wastewater Story\]](#)

- (A) 240 kg
- (B) 400 kg
- (C) 180 kg
- (D) 300 kg
- (E) 120 kg
- (F) 360 kg
- (G) 200 kg

12. Two straight lines cross. One angle measures $5x$ and the angle next to it (its linear pair) measures $4x$. Find x and the angle vertically opposite the $5x$ angle. [\[Lines & Angles\]](#)

- (A) $x = 18$, angle = 90 degrees
- (B) $x = 20$, angle = 100 degrees
- (C) $x = 20$, angle = 90 degrees
- (D) $x = 36$, angle = 100 degrees
- (E) $x = 10$, angle = 50 degrees
- (F) $x = 22$, angle = 110 degrees

13. An Amoeba captures one food particle every 24 minutes while it is active, but in a 5-hour pond observation it stays dormant for the last full hour. How many particles does it capture? [\[Nutrition in Animals\]](#)

- (A) 9 particles
- (B) 8 particles
- (C) 12 particles
- (D) 20 particles
- (E) 11 particles
- (F) 10 particles

14. Compare the two powers 2^5 and 5^2 by working out each value: state which power is bigger and the gap between them. [\[Exponents & Powers\]](#)

- (A) they are equal
- (B) 2^5 , greater by 10
- (C) 5^2 , greater by 7
- (D) 2^5 , greater by 32
- (E) 2^5 , greater by 57
- (F) 2^5 , greater by 7
- (G) 5^2 , greater by 3

15. Treated water is disinfected with chlorine at a dose of 2 mg per litre. How many grams of chlorine are needed for 5000 litres of water? [\[Wastewater Story\]](#)

- (A) 100 g
- (B) 10000 g
- (C) 2500 g
- (D) 20 g
- (E) 1 g
- (F) 2.5 g
- (G) 10 g

16. Where a slanting line crosses a pair of parallel rails, the two same-side interior angles work out to $(2x + 30)$ degrees and $(3x)$ degrees. Find the value of x . [\[Lines & Angles\]](#)

- (A) $x = 45$
- (B) $x = 90$
- (C) $x = 25$
- (D) $x = 36$
- (E) $x = 15$
- (F) $x = 30$
- (G) $x = 60$

17. A gall bladder stores 50 mL of bile to help digest fats. Each fatty meal uses 8 mL of bile. After 5 such meals, how much bile is left in the gall bladder? [\[Nutrition in Animals\]](#)

- (A) 40 mL
- (B) 18 mL
- (C) 10 mL
- (D) 2 mL
- (E) 42 mL
- (F) 0 mL

18. The Sun is about 1.5×10^8 km from Earth. Mars, at one point, is a further 0.8×10^8 km beyond that distance from Earth. Write Mars' distance from Earth in standard form. [\[Exponents & Powers\]](#)

- (A) 2.3×10^8 km
- (B) 2.3×10^9 km
- (C) 2.3×10^{16} km
- (D) 1.5×10^8 km
- (E) 23×10^7 km
- (F) 0.7×10^8 km

19. An open drain holds about 8 litres of stagnant water for every metre of its length. A colony has 250 metres of such drains. How many litres of water stagnate in them? [\[Wastewater Story\]](#)

- (A) 200 L
- (B) 4000 L
- (C) 2000 L
- (D) 258 L
- (E) 20000 L
- (F) 2008 L
- (G) 1000 L

20. Two parallel lines are cut by a transversal, making alternate interior angles of $(5y - 8)$ degrees and $(3y + 20)$ degrees. What is the size of this angle? [\[Lines & Angles\]](#)
- (A) 118 degrees
 - (B) 28 degrees
 - (C) 52 degrees
 - (D) 62 degrees
 - (E) 70 degrees
 - (F) 14 degrees
 - (G) 124 degrees
21. A patient's stomach makes 2.5 mL of acid every hour. One antacid tablet can neutralise 5 mL of this acid. For how many hours will two tablets keep the acid neutralised? [\[Nutrition in Animals\]](#)
- (A) 8 hours
 - (B) 5 hours
 - (C) 4 hours
 - (D) 10 hours
 - (E) 3 hours
 - (F) 2 hours
 - (G) 1 hour
22. Simplify $(-1)^{101} + (-1)^{200} + (-1)^0$. [\[Exponents & Powers\]](#)
- (A) 2
 - (B) 1
 - (C) -3
 - (D) 0
 - (E) 3
 - (F) -1
 - (G) -2
23. Raw sewage holds 400 mg of pollutants per litre. Primary treatment removes 30% of these, and secondary treatment then removes 80% of what is left. What is the final pollutant level? [\[Wastewater Story\]](#)
- (A) 112 mg/L
 - (B) 16 mg/L
 - (C) 44 mg/L
 - (D) 60 mg/L
 - (E) 56 mg/L
 - (F) 80 mg/L
24. Three angles lie along a straight line and measure x , $2x$ and $3x$. What is the largest of these three angles? [\[Lines & Angles\]](#)
- (A) 30 degrees
 - (B) 60 degrees
 - (C) 180 degrees
 - (D) 45 degrees
 - (E) 120 degrees
 - (F) 75 degrees
 - (G) 90 degrees

25. Food passing through a cow's four-chambered stomach spends 27 hours in the rumen, 9 hours in the reticulum, 6 hours in the omasum and 3 hours in the abomasum. What percent of the total time is spent in the rumen? [\[Nutrition in Animals\]](#)

- (A) 75%
- (B) 30%
- (C) 40%
- (D) 66.7%
- (E) 27%
- (F) 60%

26. Write 360 as a product of powers of its prime factors. [\[Exponents & Powers\]](#)

- (A) $2^4 \times 3 \times 5$
- (B) $6^2 \times 10$
- (C) $2^3 \times 3^2 \times 5$
- (D) $2^2 \times 90$
- (E) $2^3 \times 3^2 \times 5^2$
- (F) $2^3 \times 9 \times 5$
- (G) $2^2 \times 3^2 \times 10$

27. A household of 5 people uses 135 litres of water per person each day, and 80% of this becomes wastewater. How many litres of wastewater does the household produce daily? [\[Wastewater Story\]](#)

- (A) 270 L
- (B) 135 L
- (C) 675 L
- (D) 540 L
- (E) 432 L
- (F) 108 L
- (G) 600 L

28. When an angle is measured, it turns out to be double the size of whatever completes it to a right angle. How big is this angle? [\[Lines & Angles\]](#)

- (A) 20 degrees
- (B) 30 degrees
- (C) 75 degrees
- (D) 60 degrees
- (E) 40 degrees
- (F) 120 degrees

29. One gram of carbohydrate releases 4 kcal of energy and one gram of fat releases 9 kcal. A meal contains 30 g of carbohydrate and 10 g of fat. How much energy does the meal give? [\[Nutrition in Animals\]](#)

- (A) 300 kcal
- (B) 120 kcal
- (C) 210 kcal
- (D) 160 kcal
- (E) 270 kcal
- (F) 90 kcal
- (G) 250 kcal

30. Evaluate $(4^2)^3 / 4^4$ as a single power of 4, and give its value. [\[Exponents & Powers\]](#)

- (A) $4^2 = 16$
- (B) $4^4 = 256$
- (C) $4^1 = 4$
- (D) $4^2 = 8$
- (E) $4^5 = 1024$
- (F) $4^3 = 64$
- (G) 4^{24}

31. A heavily polluted pond is cleaned in 3 stages, and each stage removes half of the filth that remains. If it starts with 64 units of filth, how much filth is left after all 3 stages? [\[Wastewater Story\]](#)

- (A) 4 units
- (B) 8 units
- (C) 0 units
- (D) 32 units
- (E) 48 units
- (F) 16 units

32. Four angles meet at a point and measure 90, 85, 100 and x degrees. Find x . [\[Lines & Angles\]](#)

- (A) 90 degrees
- (B) 85 degrees
- (C) 75 degrees
- (D) 275 degrees
- (E) 95 degrees
- (F) 80 degrees

33. Along the food canal, the oesophagus is 25 cm long, the small intestine is 6 m and the large intestine is 1.5 m. What is the combined length of these three parts in centimetres? [\[Nutrition in Animals\]](#)

- (A) 750 cm
- (B) 760 cm
- (C) 700 cm
- (D) 775 cm
- (E) 850 cm
- (F) 160 cm
- (G) 625 cm

34. A sheet of paper 0.1 mm thick is folded in half 10 times, so each fold doubles the thickness. How thick is the folded stack? [\[Exponents & Powers\]](#)

- (A) 20 mm
- (B) 2 mm
- (C) 1024 mm
- (D) 102.4 mm
- (E) 1.024 mm
- (F) 10.24 mm
- (G) 51.2 mm

- 35.** Digesting sludge makes biogas: every 50 kg of sludge yields 30 cubic metres of biogas. A plant digests 2000 kg of sludge daily. How much biogas does it make each day? [\[Wastewater Story\]](#)
- (A) 2000 m³
 - (B) 60000 m³
 - (C) 2400 m³
 - (D) 12000 m³
 - (E) 600 m³
 - (F) 100 m³
 - (G) 1200 m³
- 36.** The supplement of an angle is 40 degrees more than three times the angle. What is the measure of the angle? [\[Lines & Angles\]](#)
- (A) 25 degrees
 - (B) 35 degrees
 - (C) 36 degrees
 - (D) 55 degrees
 - (E) 145 degrees
 - (F) 40 degrees
 - (G) 50 degrees
- 37.** Salivary amylase in the mouth converts 8 mg of starch into sugar every minute. A mouthful holding 200 mg of starch is chewed for 1.5 minutes. How much starch is still left unconverted when swallowed? [\[Nutrition in Animals\]](#)
- (A) 100 mg
 - (B) 160 mg
 - (C) 176 mg
 - (D) 188 mg
 - (E) 12 mg
 - (F) 194 mg
 - (G) 0 mg
- 38.** Simplify $a^7 \times a^{-3} \times a^{-2}$, where a is not zero. [\[Exponents & Powers\]](#)
- (A) a^2
 - (B) a^{42}
 - (C) a^1
 - (D) a^{-8}
 - (E) a^{12}
 - (F) a^8
 - (G) a^{-2}
- 39.** A leaking tap drips 3 mL of water every 2 seconds. How many litres of water does it waste in one full day? [\[Wastewater Story\]](#)
- (A) 129.6 L
 - (B) 12.96 L
 - (C) 130 L
 - (D) 216 L
 - (E) 64.8 L
 - (F) 259.2 L

40. Lines p and q are parallel and cut by a transversal. A pair of corresponding angles measure $(4x + 10)$ degrees and $(6x - 30)$ degrees. What is the measure of each angle? [\[Lines & Angles\]](#)
- (A) 60 degrees
 - (B) 20 degrees
 - (C) 80 degrees
 - (D) 100 degrees
 - (E) 45 degrees
 - (F) 90 degrees
41. A python eats once every 12 days, while a tiny shrew must eat 3 times a day to survive. Over 24 days, how many more meals does the shrew eat than the python? [\[Nutrition in Animals\]](#)
- (A) 2 meals
 - (B) 70 meals
 - (C) 72 meals
 - (D) 75 meals
 - (E) 68 meals
 - (F) 69 meals
42. Express 81×27 as a single power of 3. [\[Exponents & Powers\]](#)
- (A) 3^1
 - (B) 3^{10}
 - (C) 3^{12}
 - (D) 81^3
 - (E) 9^7
 - (F) 3^7
 - (G) 27^4
43. A sewer pipe is laid to slope, dropping 2 cm in height for every 5 m of length. Over a 350 m run, what is the total drop in height, in metres? [\[Wastewater Story\]](#)
- (A) 140 m
 - (B) 2.8 m
 - (C) 1.4 m
 - (D) 0.14 m
 - (E) 0.7 m
 - (F) 14 m
 - (G) 1400 m
44. What is the angle between the hour hand and the minute hand of a clock at exactly 4:20? [\[Lines & Angles\]](#)
- (A) 20 degrees
 - (B) 30 degrees
 - (C) 100 degrees
 - (D) 130 degrees
 - (E) 120 degrees
 - (F) 10 degrees
 - (G) 0 degrees

45. The large intestine reabsorbs water. Of 1.5 litres of watery food entering it, 90% of the water is reabsorbed into the body. How much water leaves in the faeces? [\[Nutrition in Animals\]](#)

- (A) 300 mL
- (B) 150 mL
- (C) 900 mL
- (D) 165 mL
- (E) 15 mL
- (F) 135 mL
- (G) 1350 mL

46. If $2^x = 64$, first find x , then work out the value of 3^x . [\[Exponents & Powers\]](#)

- (A) 64
- (B) 243
- (C) 216
- (D) 729
- (E) 6
- (F) 18

47. A plant runs 6 pumps, and each pump moves 250 litres of wastewater per minute. How many kilolitres does the plant pump in 8 hours? [\[Wastewater Story\]](#)

- (A) 720 kL
- (B) 1500 kL
- (C) 120 kL
- (D) 240 kL
- (E) 360 kL
- (F) 72 kL
- (G) 7200 kL

48. An angle of 116 degrees is bisected, and then one of the halves is bisected again. What is the measure of the smallest angle produced? [\[Lines & Angles\]](#)

- (A) 23 degrees
- (B) 14.5 degrees
- (C) 38.67 degrees
- (D) 30 degrees
- (E) 29 degrees
- (F) 58 degrees

49. A cow eats 25 kg of grass a day. Cellulose makes up 40% of the grass, and rumen bacteria digest 60% of that cellulose. How many kilograms of cellulose are digested each day? [\[Nutrition in Animals\]](#)

- (A) 10 kg
- (B) 9 kg
- (C) 4 kg
- (D) 15 kg
- (E) 6 kg
- (F) 12 kg

50. A human cell is about 2×10^{-5} m wide and a virus is about 4×10^{-8} m wide. How many times wider is the cell than the virus? [\[Exponents & Powers\]](#)

- (A) 50 times
- (B) 250 times
- (C) 5000 times
- (D) 8×10^{-13} times
- (E) 0.002 times
- (F) 500 times
- (G) 2×10^3 times

51. Dried sludge is sold as manure. A plant gets 90 kg of dry sludge from every 1000 kg of wet sludge. How many kilograms of manure come from 25 tonnes of wet sludge? [\[Wastewater Story\]](#)

- (A) 2250 kg
- (B) 900 kg
- (C) 2500 kg
- (D) 2.25 kg
- (E) 22500 kg
- (F) 225 kg
- (G) 1250 kg

52. Two angles form a linear pair and measure $(3x + 15)$ degrees and $(2x + 25)$ degrees. Find the larger of the two angles. [\[Lines & Angles\]](#)

- (A) 99 degrees
- (B) 81 degrees
- (C) 108 degrees
- (D) 72 degrees
- (E) 90 degrees
- (F) 28 degrees
- (G) 100 degrees

53. A young child has 20 milk teeth. These are later replaced by a full set of 32 permanent teeth. How many extra teeth appear by the time all the permanent teeth are present? [\[Nutrition in Animals\]](#)

- (A) 8 teeth
- (B) 20 teeth
- (C) 12 teeth
- (D) 16 teeth
- (E) 10 teeth
- (F) 52 teeth

54. Simplify $(2^2 \times 3^3)^2$ and write the answer as a product of prime powers. [\[Exponents & Powers\]](#)

- (A) $2^2 \times 3^3$
- (B) $2^4 \times 3^9$
- (C) $4^2 \times 9^3$
- (D) $2^6 \times 3^6$
- (E) $2^4 \times 6^6$
- (F) $2^4 \times 3^5$
- (G) $2^4 \times 3^6$

55. Cholera spreads through sewage-contaminated water. In a slum of 4800 people, 1 in 8 fall ill and three-quarters of those recover. How many of the ill people do NOT recover? [\[Wastewater Story\]](#)

- (A) 75 people
- (B) 150 people
- (C) 1200 people
- (D) 300 people
- (E) 200 people
- (F) 600 people

56. Find the difference between the supplement and the complement of a 50 degree angle. [\[Lines & Angles\]](#)

- (A) 170 degrees
- (B) 90 degrees
- (C) 50 degrees
- (D) 130 degrees
- (E) 40 degrees
- (F) 100 degrees

57. Food stays about 4 hours in the stomach and about 6 hours in the small intestine. If a person eats a meal at 7:30 a.m., at about what time does the food leave the small intestine? [\[Nutrition in Animals\]](#)

- (A) 6:30 p.m.
- (B) 5:30 a.m.
- (C) 11:30 a.m.
- (D) 9:30 p.m.
- (E) 1:30 p.m.
- (F) 3:30 p.m.
- (G) 5:30 p.m.

58. Among 3^4 , 4^3 and 2^6 , which is the largest? [\[Exponents & Powers\]](#)

- (A) 2^6 (128)
- (B) 4^3 (12)
- (C) 3^4 (81)
- (D) 2^6 (64)
- (E) all are equal
- (F) 3^4 (12)

59. A settling tank takes 4 hours for solids to settle and then 2 hours to drain. If it is filled at 9:45 a.m., at what time is it fully empty? [\[Wastewater Story\]](#)

- (A) 11:45 a.m.
- (B) 4:45 p.m.
- (C) 5:45 p.m.
- (D) 3:45 a.m.
- (E) 1:45 p.m.
- (F) 3:45 p.m.

- 60.** Two parallel lines are crossed by a transversal in a Z-shape. One alternate interior angle is 73 degrees. What is the measure of its co-interior (same-side interior) partner? [\[Lines & Angles\]](#)
- (A) 73 degrees
 - (B) 127 degrees
 - (C) 90 degrees
 - (D) 117 degrees
 - (E) 107 degrees
 - (F) 83 degrees
 - (G) 17 degrees
- 61.** A balanced 1800 kcal diet gives 50% of its energy from carbohydrate, 30% from fat and 20% from protein. Fat gives 9 kcal per gram. How many kilocalories come from fat, and how many grams of fat is that? [\[Nutrition in Animals\]](#)
- (A) 540 kcal, 60 g
 - (B) 360 kcal, 40 g
 - (C) 900 kcal, 100 g
 - (D) 600 kcal, 66 g
 - (E) 540 kcal, 67.5 g
 - (F) 270 kcal, 30 g
 - (G) 540 kcal, 135 g
- 62.** A single bacterium splits into two every 20 minutes. Starting from 5 bacteria, how many are there after 2 hours? [\[Exponents & Powers\]](#)
- (A) 30
 - (B) 160
 - (C) 640
 - (D) 64
 - (E) 320
 - (F) 384
 - (G) 60
- 63.** In a sample of sewage, the organic waste and inorganic waste are in the ratio 5 : 3. If the total solids weigh 4000 kg, how much of it is organic waste? [\[Wastewater Story\]](#)
- (A) 1600 kg
 - (B) 800 kg
 - (C) 2250 kg
 - (D) 2500 kg
 - (E) 3000 kg
 - (F) 2000 kg
- 64.** Two angles are supplementary, and the larger is 30 degrees less than twice the smaller. Find both angles. [\[Lines & Angles\]](#)
- (A) 50 and 130 degrees
 - (B) 75 and 105 degrees
 - (C) 45 and 135 degrees
 - (D) 80 and 100 degrees
 - (E) 60 and 120 degrees
 - (F) 70 and 110 degrees
 - (G) 70 and 100 degrees

- 65.** A dog has 42 teeth, of which 4 are long canines used for tearing flesh. What percent of the dog's teeth are canines, rounded to one decimal place? [\[Nutrition in Animals\]](#)
- (A) 10%
 - (B) 11.9%
 - (C) 12%
 - (D) 8.4%
 - (E) 4%
 - (F) 9.5%
 - (G) 9.0%
- 66.** Find the value of $5^{-2} + 2^{-2}$, written as a decimal. [\[Exponents & Powers\]](#)
- (A) 0.5
 - (B) 0.04
 - (C) 0.29
 - (D) 29
 - (E) 0.45
 - (F) 0.25
 - (G) 0.14
- 67.** A composting pit shrinks kitchen waste to one-quarter of its starting mass in 30 days. Starting from 48 kg of waste, how much mass remains and how much is lost? [\[Wastewater Story\]](#)
- (A) 12 kg remains, 12 kg lost
 - (B) 16 kg remains, 32 kg lost
 - (C) 36 kg remains, 12 kg lost
 - (D) 12 kg remains, 48 kg lost
 - (E) 6 kg remains, 42 kg lost
 - (F) 12 kg remains, 36 kg lost
 - (G) 24 kg remains, 24 kg lost
- 68.** An angle measures one-fifth of a right angle plus one-third of a straight angle. What is the size of this angle? [\[Lines & Angles\]](#)
- (A) 66 degrees
 - (B) 90 degrees
 - (C) 108 degrees
 - (D) 78 degrees
 - (E) 60 degrees
 - (F) 48 degrees
 - (G) 72 degrees
- 69.** A Hydra has 6 tentacles, and each healthy tentacle catches 2 water-fleas an hour. If one tentacle is injured and catches nothing, how many fleas does the Hydra catch in 8 hours? [\[Nutrition in Animals\]](#)
- (A) 96 fleas
 - (B) 75 fleas
 - (C) 64 fleas
 - (D) 48 fleas
 - (E) 90 fleas
 - (F) 80 fleas
 - (G) 16 fleas

70. Add 6.4×10^5 and 3.6×10^5 , and write the result in standard form. [\[Exponents & Powers\]](#)

- (A) 9.99×10^5
- (B) 10×10^{10}
- (C) 1×10^5
- (D) 10×10^5
- (E) 1×10^6
- (F) 1×10^{11}
- (G) 1×10^{10}

71. A treatment plant is designed for 10 million litres a day, but during floods it runs at 130% of this. By how many litres does the flow exceed the plant's capacity? [\[Wastewater Story\]](#)

- (A) 300,000 L
- (B) 6,000,000 L
- (C) 3,000,000 L
- (D) 13,000,000 L
- (E) 3000 L
- (F) 1,300,000 L
- (G) 30,000,000 L

72. An angle measures 110 degrees. What is the size of its reflex angle? [\[Lines & Angles\]](#)

- (A) 70 degrees
- (B) 180 degrees
- (C) 160 degrees
- (D) 220 degrees
- (E) 250 degrees
- (F) 290 degrees

73. Of 80 grams of protein eaten in a day, the body absorbs 95%. How many grams of protein pass out of the body undigested? [\[Nutrition in Animals\]](#)

- (A) 40 g
- (B) 4 g
- (C) 2 g
- (D) 8 g
- (E) 16 g
- (F) 76 g
- (G) 75 g

74. Simplify $(7^0 - 1) \times 9^{99}$. [\[Exponents & Powers\]](#)

- (A) 8
- (B) 99
- (C) 1
- (D) 9^{98}
- (E) 0
- (F) 9^{99}

- 75.** Each person produces about 0.4 kg of sewage solids per day. For a town of 25,000 people, how many tonnes of solids are produced in one week? [\[Wastewater Story\]](#)
- (A) 100 tonnes
 - (B) 140 tonnes
 - (C) 10 tonnes
 - (D) 70 tonnes
 - (E) 700 tonnes
 - (F) 7 tonnes
- 76.** Two parallel lines are cut by a transversal. One interior angle is $5x$ degrees and its alternate angle is $(x + 96)$ degrees. What is the acute angle formed at the crossing? [\[Lines & Angles\]](#)
- (A) 24 degrees
 - (B) 30 degrees
 - (C) 60 degrees
 - (D) 48 degrees
 - (E) 96 degrees
 - (F) 72 degrees
- 77.** After eating, a goat brings food back up and chews each portion of cud for 45 seconds, doing this 80 times. How many minutes of chewing is this in total? [\[Nutrition in Animals\]](#)
- (A) 36 minutes
 - (B) 60 minutes
 - (C) 90 minutes
 - (D) 3600 minutes
 - (E) 45 minutes
 - (F) 6 minutes
 - (G) 80 minutes
- 78.** A copper wire is 2.5×10^3 metres long and is sold by the centimetre. How many centimetres of wire is this, in standard form? [\[Exponents & Powers\]](#)
- (A) 2.5×10^4 cm
 - (B) 25×10^4 cm
 - (C) 2.5×10^3 cm
 - (D) 2.5×10^2 cm
 - (E) 2.5×10^5 cm
 - (F) 2.5×10^1 cm
- 79.** Treated water is reused to water a park that needs 12000 litres a day. Recycled water supplies five-sixths of this, and the rest comes from a tank. How many litres does the tank supply each day? [\[Wastewater Story\]](#)
- (A) 1000 L
 - (B) 1200 L
 - (C) 6000 L
 - (D) 10000 L
 - (E) 3000 L
 - (F) 2400 L
 - (G) 2000 L

80. An angle is 25% of its own supplement. What is the measure of the angle? [\[Lines & Angles\]](#)

- (A) 18 degrees
- (B) 144 degrees
- (C) 36 degrees
- (D) 30 degrees
- (E) 90 degrees
- (F) 60 degrees
- (G) 45 degrees

81. The liver of a 75 kg man weighs about 1.5 kg, roughly one-fiftieth of his body mass. Using the same ratio, estimate the liver mass of a 30 kg child. [\[Nutrition in Animals\]](#)

- (A) 750 g
- (B) 150 g
- (C) 300 g
- (D) 450 g
- (E) 600 g
- (F) 1500 g
- (G) 60 g

82. Simplify $(3^5 \times 3^2) / (3^3 \times 3^2)$ as a power of 3, and give its value. [\[Exponents & Powers\]](#)

- (A) 81
- (B) 9
- (C) 243
- (D) 1
- (E) 6
- (F) 3
- (G) 27

83. A grease trap catches 90% of the 50 litres of grease entering it each week, and the rest escapes into the sewer. How many litres of grease reach the sewer over 4 weeks? [\[Wastewater Story\]](#)

- (A) 180 L
- (B) 20 L
- (C) 2 L
- (D) 5 L
- (E) 200 L
- (F) 45 L
- (G) 40 L

84. Two straight lines intersect, making four angles. One angle is 7 times an adjacent angle. What is the measure of the smallest angle? [\[Lines & Angles\]](#)

- (A) 25 degrees
- (B) 22.5 degrees
- (C) 20 degrees
- (D) 30 degrees
- (E) 45 degrees
- (F) 157.5 degrees
- (G) 18 degrees

85. A tapeworm in a person's intestine absorbs 0.5 g of the host's digested food every day. Over 3 weeks, how much food does the tapeworm steal in total? [\[Nutrition in Animals\]](#)

- (A) 3.5 g
- (B) 21 g
- (C) 105 g
- (D) 7.5 g
- (E) 15 g
- (F) 10.5 g
- (G) 1.5 g

86. A square tile has a side of 10^3 mm. Find its area in square millimetres, written in standard form.

[\[Exponents & Powers\]](#)

- (A) 1×10^4 mm²
- (B) 1×10^3 mm²
- (C) 2×10^3 mm²
- (D) 2×10^6 mm²
- (E) 1×10^6 mm²
- (F) 1×10^9 mm²
- (G) 1×10^5 mm²

87. A sewage canal is 3 m wide and 1.5 m deep, and the water flows at 2 m per second. How many cubic metres of water pass a fixed point in one minute? [\[Wastewater Story\]](#)

- (A) 540 m³
- (B) 450 m³
- (C) 270 m³
- (D) 54 m³
- (E) 5400 m³
- (F) 90 m³
- (G) 9 m³

88. A transversal makes an angle of 65 degrees with one of two parallel lines. What is the sum of that angle's corresponding angle and its vertically opposite angle? [\[Lines & Angles\]](#)

- (A) 260 degrees
- (B) 230 degrees
- (C) 65 degrees
- (D) 180 degrees
- (E) 115 degrees
- (F) 130 degrees

89. An adult stomach holds about 1.2 litres when full. After a large meal it is three-quarters full, and then one-third of that food empties into the intestine. How much food, in millilitres, is left in the stomach? [\[Nutrition in Animals\]](#)

[Animals\]](#)

- (A) 300 mL
- (B) 600 mL
- (C) 450 mL
- (D) 800 mL
- (E) 400 mL
- (F) 900 mL
- (G) 200 mL

90. Find n if $2^n \times 2^3 = 2^{10}$, and then give the value of 2^n . [\[Exponents & Powers\]](#)

- (A) 2187
- (B) 1024
- (C) 64
- (D) 7
- (E) 2^{30}
- (F) 128

91. Clean water should have an oxygen demand (BOD) below 30 mg/L. A sewage sample reads 210 mg/L. What fraction of the BOD must be removed to reach the 30 mg/L limit? [\[Wastewater Story\]](#)

- (A) $\frac{6}{210}$
- (B) $\frac{7}{6}$
- (C) $\frac{1}{7}$
- (D) $\frac{30}{210}$
- (E) $\frac{5}{7}$
- (F) $\frac{6}{7}$

92. Two angles form a linear pair. At what angle do their two angle-bisectors meet? [\[Lines & Angles\]](#)

- (A) 45 degrees
- (B) 90 degrees
- (C) 180 degrees
- (D) 135 degrees
- (E) 60 degrees
- (F) 120 degrees
- (G) 30 degrees

93. Each centimetre of the small intestine carries about 200 finger-like villi. Over a 6-metre intestine, roughly how many villi are there, expressed in lakhs? [\[Nutrition in Animals\]](#)

- (A) 2 lakh
- (B) 1.2 lakh
- (C) 12 lakh
- (D) 6 lakh
- (E) 0.12 lakh
- (F) 12000
- (G) 1200

94. Write the number 50 000 000 in standard form, and state how many zeros the ordinary number has.

[\[Exponents & Powers\]](#)

- (A) 5×10^7 , 7 zeros
- (B) 5×10^6 , 6 zeros
- (C) 5×10^7 , 6 zeros
- (D) 5×10^7 , 8 zeros
- (E) 0.5×10^8 , 7 zeros
- (F) 50×10^6 , 7 zeros

- 95.** A village installs 60 low-cost toilets. Each saves 8 litres of flush water per use compared with the old type, and each toilet is used 5 times a day. How many litres of water are saved per day? [\[Wastewater Story\]](#)
- (A) 2000 L
 - (B) 4800 L
 - (C) 300 L
 - (D) 480 L
 - (E) 2400 L
 - (F) 240 L
- 96.** Angles A and B are complementary, and A is 15 degrees less than twice B. What is the measure of angle A? [\[Lines & Angles\]](#)
- (A) 25 degrees
 - (B) 55 degrees
 - (C) 65 degrees
 - (D) 60 degrees
 - (E) 45 degrees
 - (F) 35 degrees
 - (G) 50 degrees
- 97.** A thali provides 60 g of carbohydrate, 20 g of protein and 20 g of fat. Express the amounts of carbohydrate, protein and fat as a ratio in its simplest form. [\[Nutrition in Animals\]](#)
- (A) 1:1:1
 - (B) 3:2:2
 - (C) 6:2:2
 - (D) 60:20:20
 - (E) 3:1:1
 - (F) 2:1:1
 - (G) 3:1:2
- 98.** Simplify $(-2)^3 \times (-2)^2$ and give the sign and value of the answer. [\[Exponents & Powers\]](#)
- (A) -10
 - (B) 20
 - (C) 32
 - (D) -8
 - (E) -64
 - (F) -32
 - (G) -20
- 99.** Sludge is dried on sand beds, and one bed dries 200 kg of sludge in 3 days. How many bed-days are needed to dry 3 tonnes of sludge? [\[Wastewater Story\]](#)
- (A) 9 bed-days
 - (B) 450 bed-days
 - (C) 15 bed-days
 - (D) 60 bed-days
 - (E) 30 bed-days
 - (F) 45 bed-days
 - (G) 20 bed-days

100. Three angles meet around a point in the ratio 2 : 3 : 7. What is the measure of the largest angle? [[Lines](#)
& Angles]

- (A) 90 degrees
- (B) 210 degrees
- (C) 120 degrees
- (D) 105 degrees
- (E) 60 degrees
- (F) 180 degrees
- (G) 140 degrees